

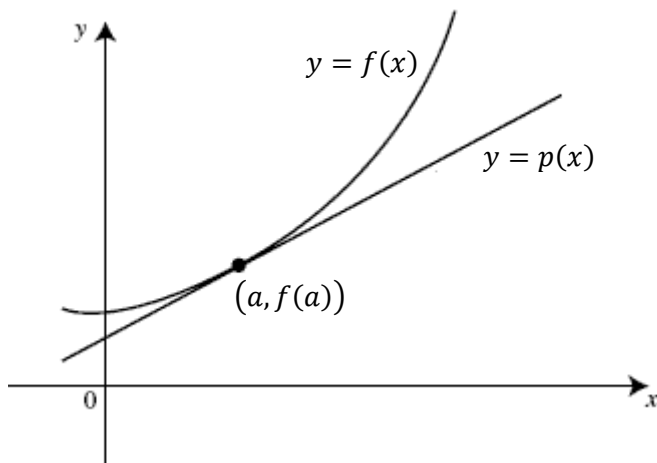
4.1 Maclaurin and Taylor Polynomials

Chapter 3 was concerned with the development of ideas surrounding the convergence or divergence of various different kinds of series. In what way is this knowledge relevant? How can it be used to provide meaningful real world results?

Recall (Local Linear Approximations):

For some functions it may be simple to evaluate $f(a)$ but difficult to compute values of f near a .

Instead we will use the equation of the tangent line at $x = a$ to estimate f near a .

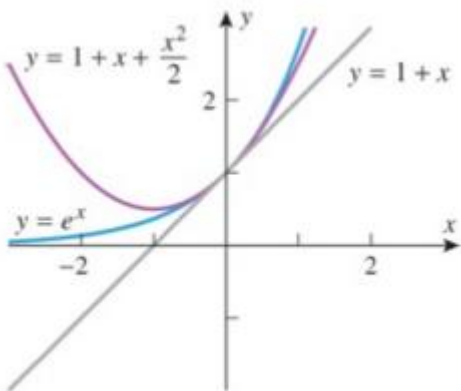


Example 1: Find the local linear approximation of $f(x) = e^x$ at $x = 0$.

Note (again): $p(x)$ only approximates $f(x)$ well when x values close to 0.
 $p(x)$ approximates $f(x)$ poorly when x values are far from 0.

The question we would like to ask is if there is any way to improve our ability to approximate values of $f(x)$.

Local Quadratic Approximations:



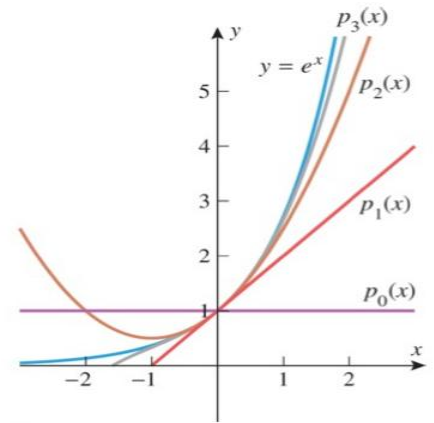
Example 2: Find the local quadratic approximation of $f(x) = e^x$ at $x = 0$.

Question: How might we be able to obtain a way to approximate values of $f(x)$ even further from $x = 0$?

Our new problem: What does the formula look like for a local approximation with a polynomial of degree n ?

Definition:

Example 3: Find the Maclaurin polynomials p_0 , p_1 , p_2 , and p_n for e^x .

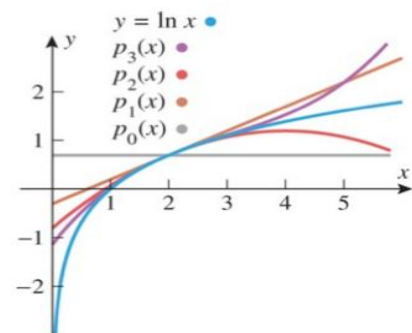


Note: Not all approximating polynomials need to be approximating a curve at $x = 0$.

We can consider approximating a curve at an arbitrary $x = x_0$.

Definition:

Example 4: Find the first four Taylor polynomial for $\ln(x)$ about $x = 2$.

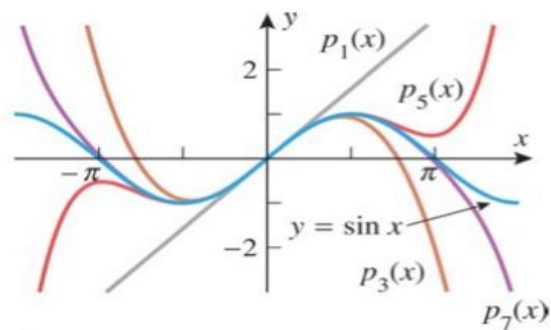


Example 5:

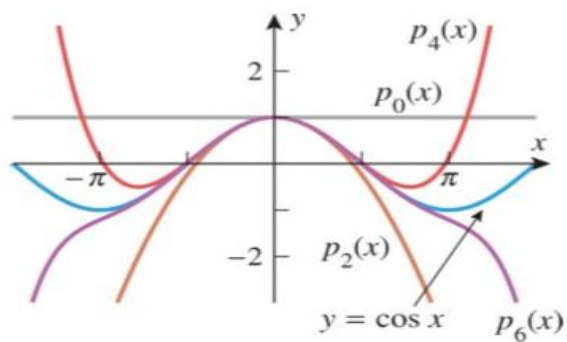
Determine the n th Taylor polynomial for $\frac{1}{x}$ about $x = 1$ and express it in sigma notation.

Example 6: Find the n th Maclaurin polynomials for ...

(a) $\sin(x)$



(b) $\cos(x)$



Example 7 (you try):

Determine the n th Taylor polynomial for...

- (a) e^{3x} about $x = 2$ and express it in sigma notation.
- (b) $\cos(x)$ about $x = \frac{\pi}{2}$ and express it in sigma notation.